

Q1. Different types of data sources in ETL: Data sources in ETL include relational databases (MySQL, Oracle), flat files (CSV, Excel), APIs, NoSQL databases (MongoDB), cloud storage (AWS S3), and ERP/CRM systems like SAP or Salesforce. These sources provide structured, semi-structured, or unstructured data for processing. Each source differs in format and access method, making extraction a critical phase in ETL.

Q2. What is data extraction? Its role in ETL: Data extraction is the first step in the ETL process where data is collected from various source systems like databases, files, or APIs. It plays a crucial role as it pulls raw data from multiple heterogeneous sources and feeds it into the pipeline for further transformation and loading. Without proper extraction, the entire ETL process cannot function correctly.

Q3. Difference between CSV and Excel in ETL: CSV is a plain text file with comma-separated values, lightweight and easy to process in ETL tools, while Excel (.xlsx) is a binary format that supports multiple sheets, formulas, and formatting. CSV is preferred in ETL for large-scale data due to its simplicity and speed, whereas Excel requires special libraries to read and is slower to process. For automated ETL pipelines, CSV is more reliable and compatible.

Q4. Steps to extract data from a relational database: First, establish a connection to the database using credentials and a connection string (JDBC/ODBC). Then write SQL queries to select the required tables and columns, apply filters if needed, and fetch the data. The extracted data is then passed to the transformation stage, and the connection is closed after extraction is complete.

Q5. Three common challenges in data extraction: The first challenge is data inconsistency, where different sources have different formats or missing values, making it hard to extract clean data. The second is handling large data volumes, as extracting millions of records can slow down the system. The third is connectivity issues, where unstable connections to APIs or databases can cause extraction failures or incomplete data pulls.

Q6. What are APIs? Role in real-time data extraction: APIs (Application Programming Interfaces) are sets of rules that allow two systems to communicate and share data over the internet using protocols like REST or SOAP. In ETL, APIs are used to extract real-time data from external platforms like Twitter, payment gateways, or weather services. They enable live data fetching without direct database access, making them ideal for real-time and cloud-based ETL pipelines.

Q7. Why databases are preferred for enterprise-level extraction: Databases are preferred because they store large volumes of structured data in an organized manner with support for complex SQL queries, indexing, and joins. They provide secure, role-based access control which is essential in enterprise environments. Additionally, databases support concurrent access, transaction management, and audit logs, making them reliable and efficient for large-scale ETL operations.

Q8. Steps for extracting data from large CSV files (1GB+): An ETL developer should use chunk-based or streaming reading techniques (like pandas read_csv with chunksize) instead of loading the entire file into memory at once. They should validate the file structure, handle missing or corrupted rows, and use parallel processing to speed up extraction. Compressed formats like GZip can also be used to reduce file size before extraction, improving overall performance.